Dhaka Crime Map — Engineering Handbook

26 Oct 2026 · repo `github.com/zawadgenweb2/dhaka-crime-map` · for engineers and coding agents

0. In five lines

- Scrapes Bangla news (Prothom Alo today), keeps likely crime stories, and has a **local LLM** (Gemma 4 E2B on the in-house GPU) extract incidents: category, exact place, date, time, evidence quote.
- Places resolve against a curated **gazetteer** (163 Dhaka places); exact spots are geocoded later via OpenStreetMap; duplicates across articles are grouped.
- **Live and Gemma-only**: incidents that pass the automatic checks are published at once (`PUBLISH_MODE=auto`); the API serves only rows from a Gemma analysis. `/review` is an optional correction tool.
- Five Docker services + one systemd model server on `genweb2inhouse`; **Postgres is the data store, job queue and lock service**.
- **No external AI APIs**: the only model is the in-house Gemma server. Dhaka is always collected; other districts are planned on demand.

1. Architecture

```
prothomalo.com ◄— collector —► Postgres 17 ◄— api (FastAPI, HTTPS :8100) ◄— browsers (LAN)
 (1 req/s)          |  scheduler      ▲  queue/locks        dashboard · /review · /docs
                    ▼                 |
              jobs table —► classifier (host network) —► llama-server (systemd, GPU, 127.0.0.1:8090)
                                       └—► OSM Nominatim (≤ 4 req/min, shared lane)
```

COMPONENT	RUNS AS	ADDRESS	OWNS
`db`	Docker `postgres:17-alpine`, volume `dhaka-crime-map_pgdata`	127.0.0.1:55432	all state, queue, advisory locks (the only copy of the data)
`migrate`	Docker, one-shot	–	schema migrations, then exits
`api`	Docker	192.168.1.200:8100 (HTTPS, self-signed)	read API, review API, "collect now" requests, static UI
`collector`	Docker, singleton	–	schedule slots, scraping, article + job rows
`classifier`	Docker, host network	–	job claims, model calls, analyses, incidents; a second thread geocodes exact spots
`dhaka-llm`	systemd `--user`	127.0.0.1:8090 + API key	llama.cpp b11669 (CUDA 12.2 build), Gemma 4 E2B Q8

Host `genweb2inhouse` (192.168.1.200): Ubuntu 20.04, RTX 3080 Ti 12 GB, NVIDIA driver 575.57 (CUDA 12.9). Code: `/home/gensIr/dhaka-crime-map`. All data lives on this server; backups: `~/dhaka-backups/` (nightly 02:30, 14 days, `dhaka-backup.timer`). Laptops hold code only. One image (`Dockerfile`) for every service; the command picks the role (`python -m app api|collector|classifier|migrate`).

2. Pipeline

#	STEP	WHERE	OUTPUT	ON FAILURE
1	Schedule 08:00/20:00 Dhaka (+ "collect now")	`collector.py`, `scheduler.py`	`schedule_slots`, `run_requests`	slot retried 3× / 10 min
2	Discover: news sitemap (48 h, titles+keywords) + daily sitemaps	`scraper.py`	candidate URLs (new only)	robots 5xx/unreachable → crawl nothing
3	URL pre-filter (no download): crime/capital sections; else crime term + Dhaka mention	`scraper.is_candidate`	≈10 of 260/day	misses are not measured yet
4	Fetch (8 MB cap, per-hop robots+host check) + full-text check	`scraper.py`, `collector.py`	`articles` (`queued`/`skipped`) + `jobs`	fetch errors counted, run continues

#	STEP	WHERE	OUTPUT	ON FAILURE
5	Classify: one call per article, JSON schema, compact prompt; place ids outside the article's shortlist → null	`worker.py`, `classifiers/gemma.py`, `classifiers/base.py`, `schema.py`	`llm_calls`, `analyses`	2 dispatches max; then `failed`
6	Validate + locate on the gazetteer place: dates, evidence quote, fallback phrase must be in the article	`store.py`, `gazetteer.py`	`incidents` (`auto` published / `failed` hidden)	failed checks are never shown
7	Group duplicates: same category + place, dates ±1, compatible time	`store.rebuild_groups`	`incidents.duplicate_of`	rule-based; can merge repeated raids
8	Exact spot (later): Nominatim, kept only ≤2.5 km from the place	`store.enrich_next`, `geocode.py`	`geocode_state` `todo` → `done` / `miss` / `far`	network error: retried later
9	Serve + map: published Gemma canonical incidents; live refresh on `data_revision`	`api.py`, `static/app.js`	heat + points + feed + progress	–
–	Optional corrections: fix category/place, reject, retry, dismiss (hides the article)	`api.py`, `static/review.html`	`review_status`	–

3. Data model (Postgres, `migrations/00*.sql`)

TABLE	PURPOSE	KEY STATES
`articles`	one row per URL (body kept for re-analysis); `analysis_version` = the only version allowed to write incidents	`queued` · `skipped` · `classified` · `failed` · `uncertain` · `dismissed`
`jobs`	work queue, one per article + analysis version; `run_id` = batch	`queued` · `running` (lease) · `done` · `failed` · `uncertain` · `cancelled` · `superseded`
`llm_calls`	every model dispatch, written **before** sending	`sent` · `ok` · `error`
`analyses`	the model's full JSON answer + tokens/cost	`materialized_at`
`incidents`	map rows	`review_status`: `auto` · `failed` · `approved` · `rejected` (`pending` only in review mode); `published`; `duplicate_of`; `geocode_state`
`data_revision`	one counter, bumped with every visible change	the dashboard reloads when it moves
`eval_reference_articles`, `eval_reference_incidents`	frozen prototype-phase labels (71 articles, 58 incidents) for offline evaluation only	never served
`runs`, `schedule_slots`, `run_requests`	collection history, one row per slot, manual triggers	slot: `pending` · `running` · `done` · `failed` · `skipped`
`geocache`, `rate_limits`	Nominatim cache and the shared request lane	–
`heartbeats`	service liveness for `/api/meta`	–
`article_reviews`, `eval_results`	reviewer verdicts, offline model evaluations	–

4. Invariants (do not break)

1. **One collector.** It holds Postgres advisory lock `7000003`; a second one waits.
2. **Claim before work.** Jobs are claimed with `FOR UPDATE SKIP LOCKED` + lease; every write checks the job `generation`.
3. **Record before calling, save before using.** `llm_calls` row before the request; `analyses` row before incidents, so a later failure never needs a second model call. A worker that dies mid-call has its job requeued when the lease expires.
4. **No network inside a DB transaction.** One short transaction replaces an article's incidents (lock `7000002`) only if the job's `analysis_version` is the article's current one; geocoding happens afterwards with conditional updates.

5. **Served = published Gemma output.** Every read path uses `db.GEMMA_PROVENANCE` (a non-legacy `gemma` analysis of the same article) plus `published`; Claude-era, imported and rules output can never be served. Every visible change calls `db.bump_revision()` in its transaction.
6. **Politeness.** robots.txt per RFC 9309 (`RobotsPolicy`), 1 req/s per publisher, Nominatim ≤ 4/min via `rate_limits`.
7. **Secrets stay out of git and images.** `.env` (0600), `deploy/tls/`, model key file; `.dockerignore` allow-lists sources only.
8. **Schema changes are new migrations**, never edits to applied ones. Saved analyses are re-read with today's schema: version breaking changes.

5. Code map

PATH	RESPONSIBILITY
`app/__main__.py`	CLI: services + `collect-now [--days N --max M]`, `reprocess-all --yes`, `stats`, `compare`, `rebuild-duplicates`
`app/reprocess.py`	re-run every collected article through Gemma (after a prompt change)
`app/config.py`	every setting (env) with defaults
`app/db.py`	connections, transactions, migrations, heartbeats
`app/scraper.py`	Prothom Alo client, sitemaps, pre-filter, article parse, robots
`app/collector.py`	collector service and `collect()`
`app/worker.py`	classifier service: claim → call → save → materialize → recover
`app/store.py`	locate, date/evidence checks, publish status, duplicate grouping, exact-spot enrichment
`app/classifiers/`	`base.py` prompts (full + compact), `gemma.py` in-house model client, `rules.py` offline baseline
`app/gazetteer.py`, `data/dhaka_places.json`	Bangla place matching (suffixes, namesake guard)
`app/api.py`, `app/static/`	REST + Swagger (incl. `POST /api/classify`: try a pasted article, nothing saved), dashboard, review page
`app/evaluate.py`, `scripts/eval_table.py`	offline model evaluation vs the frozen 71-article reference set
`deploy/inhouse-llm/`	llama.cpp build, model download, start scripts, systemd units, vLLM setup
`compose.yaml`, `compose.inhouse.yaml`	services; server overrides (HTTPS, loopback DB, host-network classifier)

6. Configuration (`.env`, see `.env.example`)

`POSTGRES_PASSWORD` · `ADMIN_TOKEN` (review + token-triggered runs) · `API_BIND` (server LAN IP) · `CLASSIFIER` (`gemma` | `rules`) · `GEMMA_API_KEY`, `GEMMA_BASE_URL`, `GEMMA_MODEL`, `GEMMA_PROMPT` (`compact` | `full`) · `MODEL_LABEL` · `PUBLISH_MODE` (`auto` default | `review`) · `SCRAPE_TIMES` · `USER_AGENT` · `MAX_ARTICLES_PER_RUN` · `GEOCODE_*` · `ALLOW_UI_TRIGGER` · `GOOGLE_MAPS_API_KEY` (browser key; empty = OpenStreetMap tiles).

7. Runbook

```
ssh genweb2inhouse && cd ~/dhaka-crime-map
C="docker compose -f compose.yaml -f compose.inhouse.yaml"
git pull && $C up -d --build           # deploy (migrations run automatically)
$C ps ; $C logs -f collector classifier # health and logs
$C exec classifier python -m app collect-now      # scrape now
$C run --rm migrate collect-now --days 90 --max 3000   # backfill 90 days (progress shows on the dashboard)
$C stop classifier && $C run --rm migrate reprocess-all --yes && $C up -d   # re-run all articles after a prompt
$C exec classifier python -m app stats
$C exec db psql -U dcm -d dcm          # SQL shell
systemctl --user status|restart dhaka-llm ; journalctl --user -u dhaka-llm -f   # model server
$C --profile test build test && $C --profile test run --rm test   # 66 tests, separate dcm_test database
deploy/backup-db.sh                    # backup now; restore: pg_restore -U dcm -d dcm --clean <file>
```

SYMPTOM	CHECK	FIX
Run `failed: robots.txt disallows …`	publisher reachable?	transient: slot retries; persistent: network or publisher block
Jobs stay `queued`	`curl -s 127.0.0.1:8090/health`; classifier logs	`systemctl --user restart dhaka-llm`
Classifier log "pausing 10 min"	401/404 from model server	key in `.env` vs `~/.config/dhaka-llm/api-key`; model alias `dhaka-gemma`
Map not updating	`/api/meta` `data_version`, `batches`, classifier logs	open pages poll every 5 s while work runs, 30 s otherwise
Browser "Not secure"	self-signed LAN cert	Tailscale HTTPS (needs tailnet admin) or a domain + Let's Encrypt
GPU errors after kernel update	`nvidia-smi`, `dkms status`	DKMS must rebuild driver 575 for the new kernel

8. Adding features (recipes)

- **New model:** download a GGUF into `~/.local/share/dhaka-llm`, set `DHAKA_LLM_MODEL` via `systemctl --user edit dhaka-llm`, restart, then `$C exec classifier python -m app compare --with gemma --limit 71 --out /tmp/r.json`; compare with the table in §9, then `reprocess-all`.
- **New newspaper (planned design):** source registry + per-source config (sitemaps, article URL pattern, crime/capital sections, body selector); probe robots, bot challenges and extraction before enabling. Samakal fits (same sitemap format, body in `#contentDetails`); Bangladesh Pratidin returns a Cloudflare challenge (needs publisher allowlisting).
- **Other districts (planned):** national gazetteer from `data/sources/wikidata-bd-admin-*.json` (CC0); ambiguity-aware matching; coverage table + token-protected activation + resumable 30-day backfill; schema v2 replaces `in_dhaka`.
- **New category:** `schema.py` (enum + labels), prompt rules in `classifiers/base.py`, colours in `static/styles.css` / `app.js`; re-evaluate.
- **Schema change:** add `migrations/00N_*.sql`; never edit applied ones; add a service test in `tests/test_services.py`.
- **API endpoint:** in `create_app()`; work-triggering or private endpoints use `Security(ADMIN_KEY)`; add a test; check `/docs`.

9. Tests and evaluation

Tests: 66 (pure + the separate `dcm_test` database, never `dcm`). Evaluation reference: 71 articles labelled during the prototype phase, frozen in `eval_reference_*` (a strong baseline, not ground truth). E2B, compact prompt + calendar: JSON 100%, article P/R 0.86/0.88, incident P/R 0.76/0.71, category 0.78, thana 0.93, date 0.53, 2.4 s/article. The gates in `evaluate.GATES` are a yardstick, not a publishing switch.

Model/engine benchmark, 26 Oct (71 articles, same prompt, RTX 3080 Ti; `GEMMA_STREAM=1 ... compare` measures speed; ±0.03 run-to-run):

SETUP	INCIDENT P / R	ARTICLE P / R	TIME/ARTICLE P50	PREFILL	DECODE	GPU MEM
E2B Q8, llama.cpp (live)	0.80–0.83 / 0.64–0.66	0.87–0.90 / 0.81	2.4 s	~4,400 tok/s	~120 tok/s	2.9 GB
E4B QAT Q4_0, llama.cpp	0.71 / 0.69	0.78 / 0.83	2.8 s	~3,100 tok/s	~113 tok/s	3.5 GB
E4B QAT w4a16, vLLM 0.32	stopped (see below)		~13 s (long pretty JSON)		~120 tok/s	11.7 of 12 GB

vLLM lessons on 12 GB (kept in `deploy/inhouse-llm/start-vllm.sh`, not in use): the w4a16 checkpoints keep the per-layer embeddings in bf16 (E4B 11.5 GB, E2B 8.35 GB on disk; E4B uses 10.1 GB loaded); torch.compile autotuning runs out of memory, eager mode decodes at ~15 tok/s, decode-only CUDA graphs (`-cc mode 0, FULL_DECODE_ONLY`) restore ~120 tok/s; xgrammar's unbounded whitespace makes the model loop on newlines and its no-whitespace mode breaks answers (recall 0.02), so use the `guidance` backend with `GEMMA_JSON_WHITESPACE=bounded`. Decision: stay on llama.cpp E2B.

10. Decisions

DECISION	WHY
Postgres as queue/locks, no broker	small volume; one fewer service; transactional claims

DECISION	WHY
llama.cpp built for CUDA 12.2, outside Docker	no NVIDIA container toolkit on the shared host; driver-compatible
vLLM installed (cu129 wheel) but not serving	default PyPI wheel needs CUDA 13; speed benchmark pending
Compact prompt + grammar-constrained `place_id`	thana accuracy 71% → 93–98% for the small model
Auto-publish, review optional (26 Oct)	product decision: articles state crimes plainly; live map over a review backlog. Evidence check proves a quote exists, not that it supports the claim
llama.cpp, not vLLM (26 Oct)	same speed for one-at-a-time requests; vLLM's QAT checkpoints need most of the 12 GB card and long JSON output
Geocode after publishing	Nominatim allows 1 request / 15 s; inline geocoding would stall the GPU for hours in a backfill
Driver 535 → 575, no OS upgrade	enough for current vLLM/llama.cpp; OS upgrade unsafe without console access

11. Known gaps

Backups stay on the same machine (copy off-site) · duplicate rule can merge repeated raids at one place (Dayaganj 19/20 Oct) · re-classifying an article drops its review corrections · daily sitemaps have no titles, so backfill covers only the crime/capital sections (older Savar/Keraniganj district news is missed) · Gemma can still split one episode into several incidents · pre-filter recall unmeasured · `/api/meta` exposes operational data (fine on LAN, not public) · Tailscale HTTPS needs the tailnet admin · server: Redis on all interfaces, NTP unsynchronised, `sign2text` restart loop (not ours).

12. Notes for coding agents

- Commit as **zawadgenweb2** (repo-local git config); **never add AI co-author or "generated by" lines**.
- Never print or commit secrets (`.env`, `deploy/tls/key.pem`, `~/.config/dhaka-llm/api-key`).
- Run the test suite in Docker before pushing; deploy with the runbook commands; keep invariants in §4.
- Design changes get an independent Codex council review first (model `gpt-6.1-sol`, effort `medium`).